ENHANCED SPLITWISE: AN INTELLIGENT EXPENSE MANAGEMENT AND GROUP SETTLEMENT PLATFORM

Design, Implementation, and Evaluation of a Multi-Tier Financial Architecture with Computer Vision, Language Model Routing, and Deterministic Settlement

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ABSTRACT

Personal and group financial management remains hindered by friction across transaction capture, multi-party bill division, categorization, outlier detection, and future outlay planning. Existing platforms either rely entirely on manual user inputs or attempt to apply generative artificial intelligence directly to mathematical calculations, leading to rounding errors and arithmetic hallucinations.

This project presents **Enhanced Splitwise (`enhanced-sw`)**, a multi-tier financial management platform that combines computer vision, large language model (LLM) orchestration, and deterministic mathematical algorithms. The system employs a multi-tiered architecture comprising a cross-platform client (React Native / Expo with custom application navigation), an application gateway (Node.js / Express / TypeScript), a relational persistence layer (PostgreSQL), and an AI microservice (Python / FastAPI).

The platform introduces five core technical capabilities:

- **Hybrid Perception-Calculation Pipeline**: Local optical character recognition (PaddleOCR) with vertical and horizontal reading-order sorting and multi-provider LLM structured parsing (`gemini-3.6-flash`, `mistral-small-latest`, and `openai/gpt-oss-20b` via Groq), coupled with deterministic fixed-point arithmetic (`Decimal`) to ensure exact receipt balance reconciliation down to individual cents/paise ($\sum S_j = Total$).
- **Leave-One-Out (LOO) Statistical Anomaly Detection**: Category-level outlier detection using dynamic statistical thresholding that excludes the target transaction from baseline mean and variance estimation to eliminate outlier masking effects.
- **Greedy Debt Simplification**: A bipartite heap-based graph optimization algorithm that reduces N-participant pairwise debt networks into a compact settlement plan with at most $N-1$ direct cash transfers.
- **Time-Series Spending Forecasting**: Meta Prophet Generalized Additive Models (GAM) modeling linear trend, weekly seasonality, and 80% uncertainty intervals over zero-filled daily expenditure series, featuring an adaptive moving-average cold-start fallback.
- **Decoupled Spending Insights**: Deterministic statistical aggregations (burn rate, category variance, budget adherence) fed as structured parameters into LLMs for qualitative commentary without delegating mathematical computations.

Automated testing confirms system reliability across 109 passing Python unit/integration tests (6 mock tests skipped, 0 failures), 48 passing backend API tests across 9 test suites (0 failures), and 0 static TypeScript compilation errors. The system was validated against realistic datasets and representative interface workflows.

Keywords: Expense Management, Optical Character Recognition, Large Language Models, Deterministic Arithmetic, Debt Simplification, Anomaly Detection, Meta Prophet Forecasting.

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SECTION 1 — INTRODUCTION

1.1 Background

Personal financial management is an essential factor in individual economic well-being. With the growth of digital payments, cashless transactions, and shared living expenses (such as shared apartments, group travel, and dining), individuals frequently manage multiple unstructured financial transactions each week. Despite the availability of smartphones, many consumers still record expenditures manually or rely on disconnected tools that address only single parts of personal finance.

Existing expense tracking applications generally fall into two categories:

- **Single-user personal budgeting apps** (e.g., Mint, Spendee, PocketGuard): These focus on personal budget limits and bill reminders, but lack itemized multi-person bill splitting and shared group settlement features.
- **Group debt utilities** (e.g., standard Splitwise, Tricount): These calculate group balances, but typically do not provide automated receipt OCR parsing, semantic transaction categorization, statistical anomaly detection, or predictive spending forecasts.

Recent efforts to apply Generative Artificial Intelligence (AI) to financial tools often introduce significant failure modes: Large Language Models (LLMs) are prone to arithmetic errors, hallucinated amounts, and rounding inconsistencies when asked to calculate item shares, taxes, and group balances. As a result, an effective system must combine computer vision and language models for unstructured data *perception*, while delegating all arithmetic calculations to deterministic algorithms.

1.2 Need / Motivation

The development of **Enhanced Splitwise (`enhanced-sw`)** addresses eight practical challenges in personal and group financial management:

- **Manual Entry Effort and Errors**: Manually typing itemized paper receipts into mobile applications takes time, leads to transcription mistakes, and discourages consistent tracking.
- **Complex Multi-Item Bill Splitting**: Restaurant and retail receipts often include shared starters, individual mains, taxes, service charges, and discounts. Dividing these fairly without automated line-item assignment is tedious.
- **Discrepancy and Penny Imbalance**: Independent rounding across multiple shares often leaves a small leftover amount ($\Delta \neq 0.00$), creating confusion during group settlement.
- **Unstructured Categorization**: Categorizing expenditures by hand requires continuous user effort. Simple keyword matching often fails on cryptic merchant names.
- **Group Debt Growth**: In groups where members pay for each other reciprocally, pairwise debts increase quadratically ($O(N^2)$), creating unnecessary transactions and settlement complexity.
- **Lack of Outlier Awareness**: Standard expense trackers do not warn users when an individual purchase is unusually high compared to typical spending in that category.
- **Limited Forward Visibility**: Most tools only show historical summaries (such as past expense pie charts) without providing spending trajectory forecasts for the remainder of the month.
- **LLM Calculation Inaccuracies**: Using conversational AI assistants to calculate shared bills often results in incorrect math, violating the basic requirement of financial applications for exact accounting.

1.3 Problem Statement

To design, develop, and evaluate an integrated financial management and group settlement platform that:

- Ingests receipt images using Optical Character Recognition (OCR) and structured entity extraction;
- Provides deterministic cent-level reconciliation for multi-participant itemized splits;
- Simplifies group debt networks into a compact settlement plan with at most $N-1$ transfers using greedy graph algorithms;
- Identifies unusual spending anomalies using statistical category baselines;
- Forecasts future expenditures using additive time-series models with uncertainty intervals; and
- Generates contextual financial insights while preventing LLM calculation errors.

1.4 Objectives

To achieve this problem statement, the project establishes six concrete technical objectives:

- **Multi-Tier Architecture**: Implement a modular, cross-platform architecture separating presentation (React Native / Expo with custom navigation), business routing and persistence (Node.js / Express / PostgreSQL), and specialized AI/ML processing (Python / FastAPI).
- **Multi-Provider OCR & Extraction Pipeline**: Build an OCR pipeline leveraging PaddleOCR for bounding box detection and reading-order layout sorting, orchestrated with a multi-model LLM router (`gemini-3.6-flash` → `mistral-small-latest` → `openai/gpt-oss-20b`) using Pydantic JSON schemas.

- **Deterministic Financial Math Engine**: Implement a fixed-point `Decimal` accounting engine that handles proportional tax/discount distribution and penny rounding reconciliation down to the smallest currency unit.
- **Bipartite Debt Simplification Engine**: Implement a greedy graph algorithm that reduces complex group liability networks to at most *N-1* settlement transfers.
- **Leave-One-Out Anomaly Detection Engine**: Implement dynamic category-based outlier detection that recalculates historical baseline parameters excluding the candidate transaction to avoid masking.
- **Time-Series Forecasting Engine**: Integrate Meta Prophet to decompose historical daily spending into trend and weekly seasonality, providing 30-day projection bounds, 80% uncertainty intervals, and an adaptive cold-start fallback.

1.5 Technical Contributions

The main engineering contributions of `enhanced-sw` are:

- **Separation of AI Perception and Deterministic Math**: A design where LLMs are restricted to token extraction, layout understanding, and qualitative natural language commentary, while all currency calculations are executed by verified fixed-point algorithms.
- **Penny-Reconciled Pro-Rata Tax Allocation**: A closed-form mathematical pipeline for distributing itemized service charges, tips, and taxes proportionally across participants based on their pre-tax subtotals, with residual penny balancing.
- **Masking-Resistant Outlier Detection**: Implementation of a Leave-One-Out (LOO) statistical formulation for transaction anomaly flagging, preventing large outliers from distorting the sample mean and variance.
- **Multi-Tier LLM Router with Provider Fallback**: An automated failover mechanism that improves resilience against transient provider failures and API rate limits by cycling across heterogeneous model APIs while enforcing identical Pydantic schema validation.
- **Continuous Time-Series Resampling for Financial GAMs**: A financial preprocessing pipeline that converts sporadic, sparse expense records into continuous, zero-filled daily series suitable for Meta Prophet forecasting with automatic sample-size gating ($N \geq 5$).

1.6 System Overview

The `enhanced-sw` platform operates across four coordinated tiers:

- **Presentation Tier**: Cross-platform client application built with React Native and Expo, providing responsive screens for transaction entry, receipt review, budget tracking, debt settlement graphs, and analytical dashboards.
- **Application Gateway Tier**: REST API gateway written in TypeScript and Node.js (Express), managing client routing, request validation, database interactions, and secure proxying to the Python AI microservice.
- **Data Tier**: Relational PostgreSQL persistence layer storing expenses and monthly budget configurations, supported by an in-memory repository fallback for local development and testing. Group settlement is evaluated statelessly from client-supplied transaction arrays.
- **AI & Analytics Tier**: Python 3.12 FastAPI microservice hosting PaddleOCR, LangChain LLM routers, the `Decimal` math engine, the greedy settlement solver, the LOO anomaly detector, and the Meta Prophet time-series engine.

1.7 Architectural Diagrams

The system architecture and specialized algorithmic pipelines are illustrated in Figures 1.1 through 1.6:

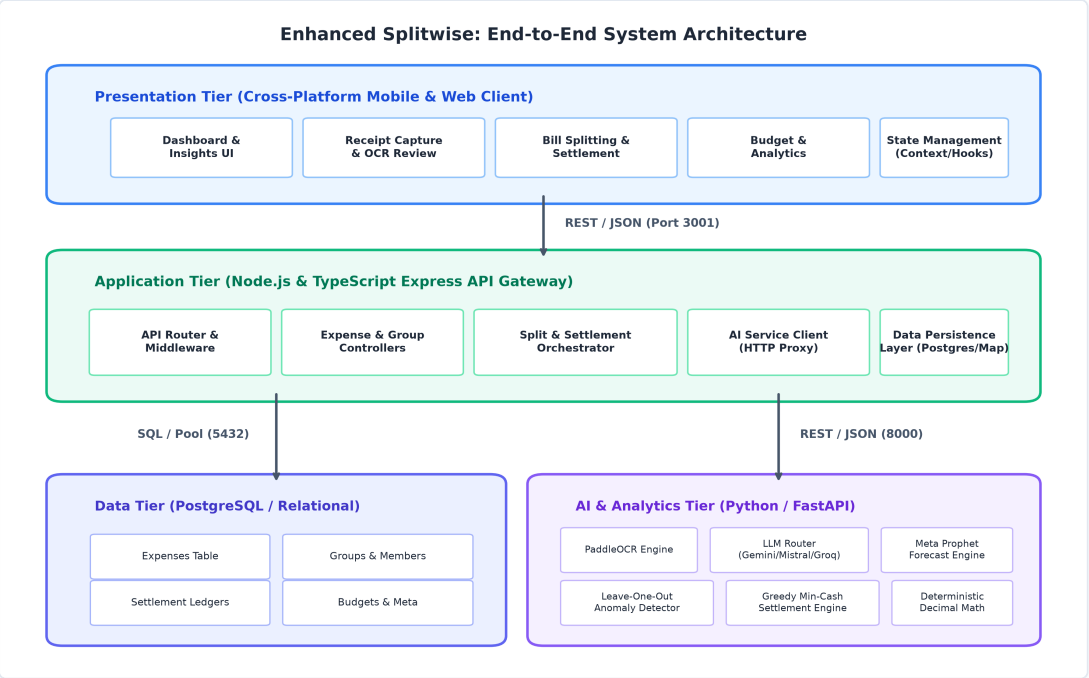


Figure 1.1: Multi-tiered architecture of the Enhanced Splitwise platform showing Presentation, Application, Data, and AI/ML tiers.

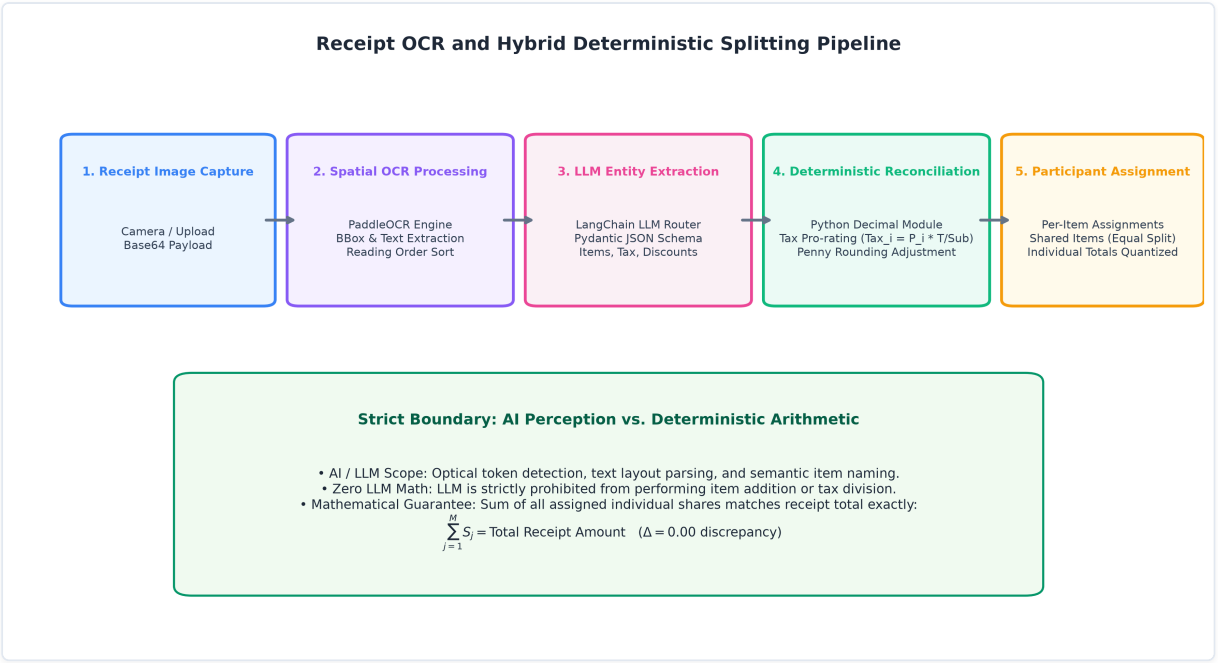


Figure 1.2: Hybrid bill splitting pipeline showing strict boundary between LLM entity perception and deterministic Decimal reconciliation.

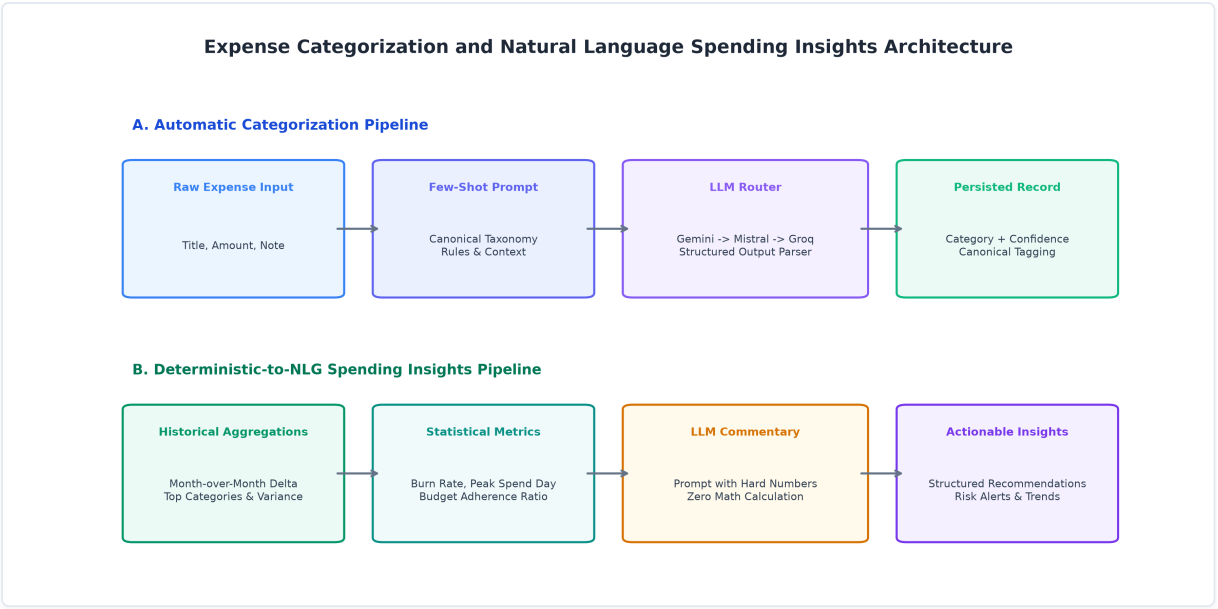


Figure 1.3: Dual architecture for automatic transaction classification and deterministic-to-NLG spending insight generation.

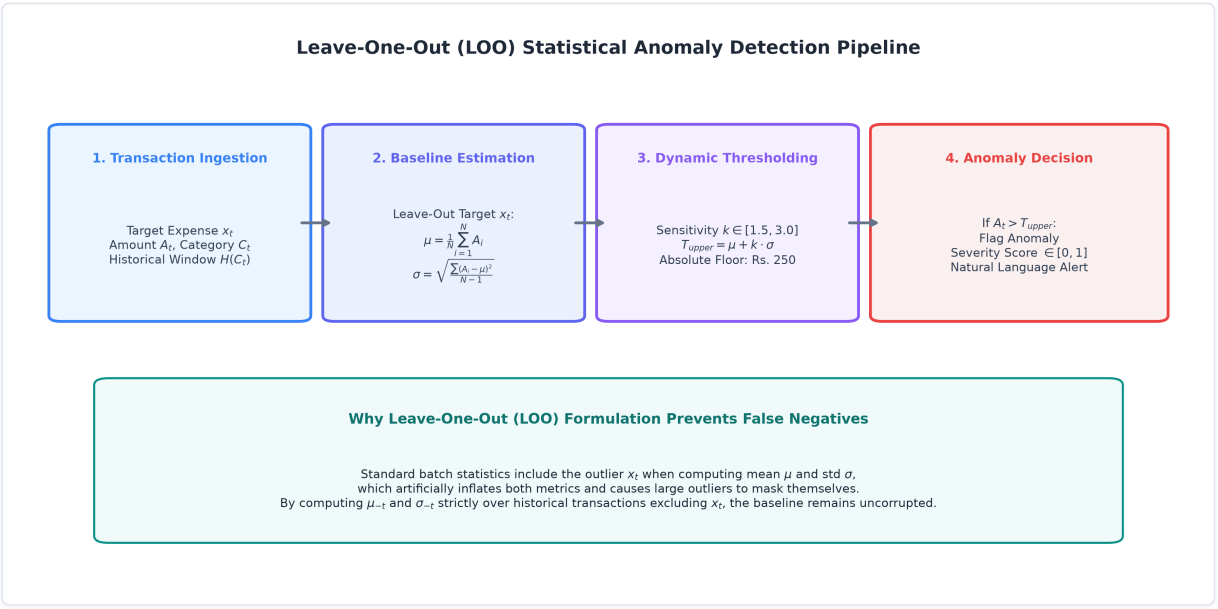


Figure 1.4: Leave-One-Out (LOO) anomaly detection pipeline detailing dynamic thresholding and outlier masking prevention.

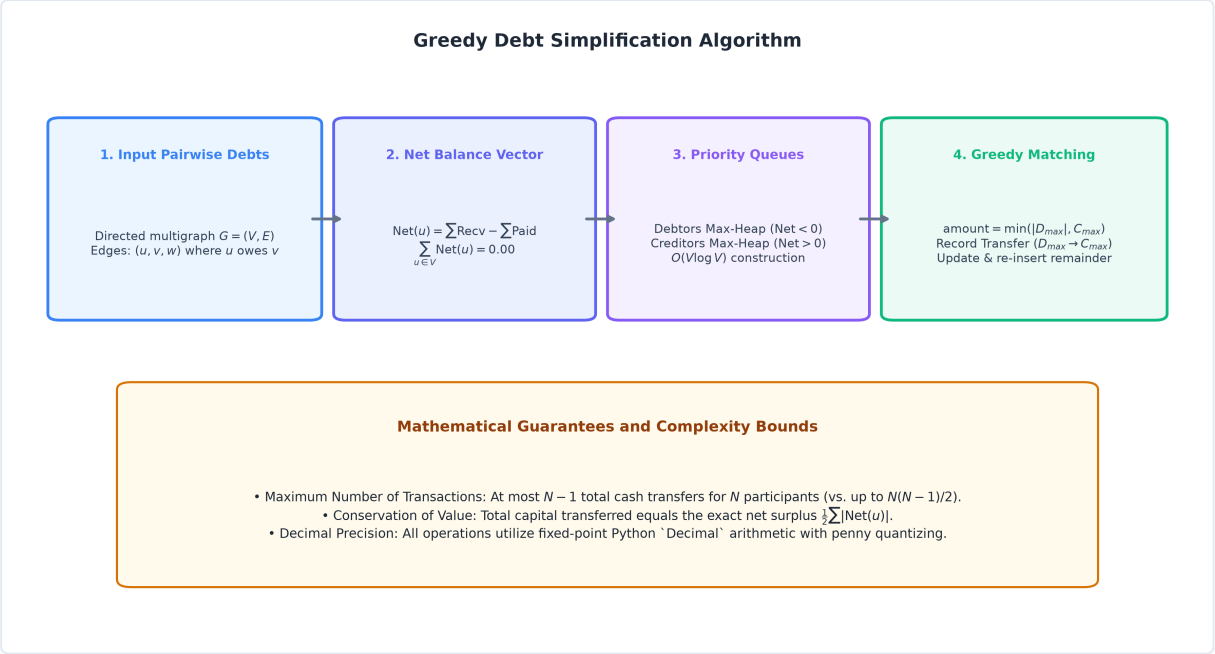


Figure 1.5: Greedy Debt Simplification Algorithm showing net balance extraction, max-heaps, and greedy settlement matching.

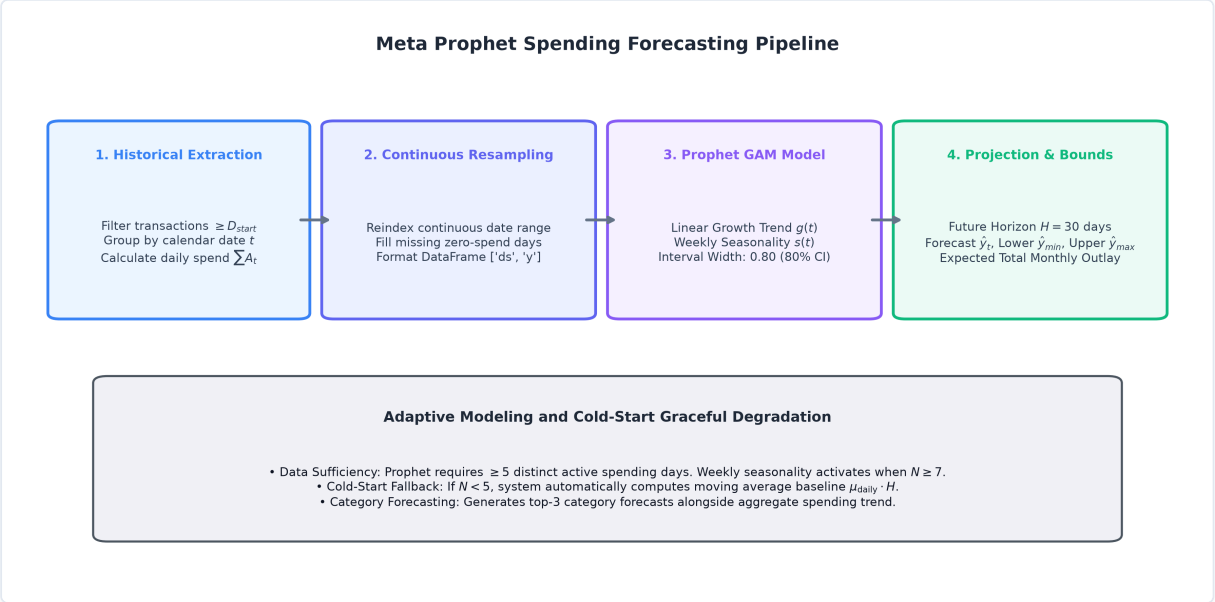


Figure 1.6: Time-series expenditure forecasting pipeline detailing continuous daily resampling, GAM decomposition, and fallback gating.

SECTION 2 — LITERATURE SURVEY

2.1 Thematic Literature Survey

To position `enhanced-sw` within contemporary research, we examine 20 selected academic and research sources across six core domains: Optical Character Recognition and Receipt Extraction, Time-Series Spending Forecasting, Graph Debt Simplification, Financial Anomaly Detection, In-Context Learning and Transaction Categorization, and Tool Augmentation.

A. Receipt OCR and Layout-Aware Information Extraction

Early receipt processing relied on standalone OCR pipelines and heuristic engines. Huang et al. (2022) developed LayoutLMv3, demonstrating that combining text, visual, and spatial coordinate embeddings improves key-value extraction on complex documents. Du et al. (2020) presented PP-OCR in an arXiv preprint, achieving lightweight CPU text detection and recognition. Jaume et al. (2019) introduced the FUNSD benchmark for spatial form understanding, while Garncarek et al. (2021) demonstrated in LAMBERT that relative 2D coordinate embeddings in transformer self-attention enhance layout-aware extraction. Carbonell et al. (2019) explored tree representations

for structured document analysis, and Xu et al. (2020) introduced LayoutLM for multimodal document pre-training. `enhanced-sw` builds upon these insights by pairing lightweight local PaddleOCR with a deterministic reading-order spatial sorter and structured LLM extraction.

B. Financial Time-Series Forecasting

Forecasting personal expenditures is challenging due to irregular transaction frequencies and seasonal variance. Taylor and Letham (2018) introduced Meta Prophet, an additive regression model combining non-linear trends, Fourier-based seasonality, and holiday effects. Lim and Zohren (2021) surveyed deep learning architectures for time series, noting that while deep models excel on large datasets, they require substantial training data. Salinas et al. (2020) developed DeepAR for probabilistic autoregressive forecasting across related series. Hyndman and Athanasopoulos (2018) established principles for evaluating seasonal patterns and baseline benchmarks. `enhanced-sw` adopts Prophet with continuous zero-filled daily series resampling, 80% uncertainty intervals, and an adaptive moving-average fallback.

C. Graph-Theoretic Debt Simplification and Network Flows

Multilateral obligation clearing has foundations in mathematical economics and network optimization. Eisenberg and Noe (2001) established clearing payment vectors in financial networks, proving existence and uniqueness under systemic balance constraints. Schuldenzucker et al. (2020) analyzed default clearing tractability, noting that linear cash-flow networks remain computationally tractable. Kavitha et al. (2009) surveyed cycle bases and cycle cancellation algorithms in directed graphs, while Ahuja, Magnanti, and Orlin (1993) established classic network flow algorithms and conservation invariants. `enhanced-sw` builds upon network clearing theory by implementing a greedy bipartite heap-matching algorithm that reduces *N*-participant debt networks to a compact settlement plan with at most *N-1* direct transfers.

D. Statistical and Contextual Anomaly Detection

Financial outlier detection has traditionally focused on fraud and unusual activity. Chandola et al. (2009) provided a foundational taxonomy of point, contextual, and collective anomalies. Ahmed et al. (2016) surveyed anomaly detection in financial data, highlighting that statistical distance metrics provide strong explainability. Chalapathy and Chawla (2019) surveyed deep anomaly detection in an arXiv preprint, noting that deep models can lack transparent reasoning. Thudumu et al. (2020) analyzed outlier techniques for high-dimensional data. `enhanced-sw` implements a Leave-One-Out (LOO) statistical formulation that prevents candidate outliers from corrupting historical mean and variance baselines.

E. In-Context Learning and Transaction Categorization

Transaction classification in modern financial interfaces benefits from zero-shot and few-shot natural language understanding. Brown et al. (2020) demonstrated in their seminal work on GPT-3 that large language models perform robust few-shot and zero-shot categorization across unseen text distributions when guided by structured context. `enhanced-sw` leverages this in-context classification capability across 11 canonical financial categories with strict Pydantic schema validation.

F. Tool Augmentation and Decoupled LLM Reasoning

Schick et al. (2023) introduced Toolformer, demonstrating that teaching language models to invoke external tools for mathematical and factual operations significantly reduces hallucinations. This principle is foundational to `enhanced-sw`: the LLM is restricted to semantic extraction and textual structuring, while all arithmetic calculations are executed by deterministic code.

2.2 Comparative Literature Table (20 Sources)

Table 2.1 presents a systematic comparative analysis of the 20 selected academic and research sources foundational to this project.

#	Paper Title & Authors	Year	Publication Venue	Problem Addressed	Methodology / Technique	Key Contribution	Identified Limitation / Gap	Relevance to Enhanced-SW
1	**LayoutLMv3: Pre-training for Document AI** *Y. Huang et al.*	2022	ACM MM '22	Multimodal document understanding across vision, layout, and text	Unified masked text/image transformer with patch alignment	High accuracy on document benchmarks without heavy CNN backbones	High inference resource footprint; no arithmetic reconciliation	Informs spatial layout parsing while keeping calculations in deterministic layers
2	**PP-OCR: Practical Ultra Lightweight OCR** *Y. Du et al.*	2020	arXiv:2009.09941 (Preprint)	Low-latency text detection and recognition on CPU/edge	DBNet text detection + SVTR text recognition (<10MB size)	Efficient CPU text recognition suitable for real-time processing	Raw tokens lack semantic entity relationship understanding	Deployed directly as primary OCR engine in `ocr_service.py`
3	**FUNSD: Form Understanding Dataset** *G. Jaume et al.*	2019	IEEE ICDARW '19	Benchmark for spatial form understanding on noisy documents	Scanned forms with word-level spatial coordinates and links	Standardized benchmark for evaluating spatial relation extraction	Focuses on administrative forms rather than retail thermal receipts	Empirical baseline for handling OCR token grouping and spatial noise
4	**LAMBERT: Layout-Aware Language Modeling** *Ł. Garncarek et al.*	2021	EACL '21	Integrating 2D coordinate embeddings into Transformer attention	Relative spatial coordinates embedded into query-key attention	Proved spatial coordinates improve receipt information extraction	Computationally heavy; does not perform consumer share allocation	Validates spatial reading-order sorting heuristic (`_sort_reading_order`)
5	**Information Extraction using Tree-Representation** *M. Carbonell et al.*	2019	IEEE ICDAR '19	Extracting hierarchical line-item structures from receipts	Tree-structured parsing and graph convolutional networks	Captured nested item-to-subtotal relationships across layouts	Brittle against thermal paper distortions and unusual layouts	Replaced with zero-shot LLM semantic extraction with Pydantic schemas
6	**LayoutLM: Pre-training of Text and Layout** *Y. Xu et al.*	2020	ACM KDD '20	2D spatial layout and text pre-training for document AI	Joint modeling of text position coordinates and visual features	Seminal document transformer demonstrating 2D positional gains	Requires extensive pre-training; lacks balance reconciliation	Informs spatial coordinate preprocessing prior to LLM schema parsing
7	**Forecasting at Scale** *S. J. Taylor & B. Letham*	2018	The American Statistician	Automated time-series forecasting robust to trend changes	Generalized Additive Model (GAM) with trend + seasonality	Robust decomposable time-series model (Meta Prophet)	Requires continuous daily series; degrades on sparse intervals	Core engine of `SpendingForecasterService` with zero-filled daily series
8	**Time-Series Forecasting with Deep Learning** *B. Lim & S. Zohren*	2021	Phil. Trans. R. Soc. A	Survey of deep neural forecasting architectures	Systematic review of RNNs, TCNs, and Temporal Fusion Transformers	Benchmark of scenarios where deep networks outperform statistics	Deep models require thousands of samples; overfit on sparse data	Justifies using Prophet over heavy deep networks for sparse expense data
9	**DeepAR: Probabilistic Forecasting** *D. Salinas et al.*	2020	Int. J. Forecasting	Probabilistic forecasting with predictive uncertainty intervals	Autoregressive LSTM estimating parametric distributions	High-quality quantile forecasts across related time series	Severe degradation when trained on small isolated individual series	Informs Prophet's additive Bayesian 80% uncertainty interval design
10	**Forecasting: Principles and Practice** *R. J. Hyndman & G. Athanasopoulos*	2018	Textbook (OTexts)	Methodological principles for time-series decomposition	Classical decomposition, seasonal adjustments, and baseline metrics	Established evaluation benchmarks for seasonal time series	Broad text without automatic cloud API failover	Informs moving-average baseline fallback for short histories ($N < 5$)
11	**Systemic Risk in Financial Systems** *L. Eisenberg & T. H. Noe*	2001	Management Science	Multilateral clearing in interconnected payment networks	Fixed-point vector lattice formulation and clearing	Proved existence and uniqueness of clearing	Focuses on institutional default rather than peer transfer plans	Theoretical basis for net balance conservation ($\sum Net = 0$)

#	Paper Title & Authors	Year	Publication Venue	Problem Addressed	Methodology / Technique	Key Contribution	Identified Limitation / Gap	Relevance to Enhanced-SW
					payment matrix	payment vectors		
12	**Default Ambiguity in Financial Networks** *S. Schuldenzucker et al.*	2020	Operations Research	Clearing tractability in financial debt graphs	Algorithmic game theory and computational complexity analysis	Proved linear cash-flow networks remain computationally tractable	Does not construct human-readable peer settlement graphs	Guides linear net-balance reduction before greedy transfer matching
13	**Cycle Bases in Graphs** *T. Kavitha et al.*	2009	Computer Science Review	Minimum cycle bases and debt cancellation in graphs	Algorithmic review of cycle cancellation algorithms	Established methods for eliminating circular indebtedness in graphs	Cycle cancellation leaves linear multi-hop chains unsettled	Extended to greedy bipartite matching achieving at most $N-1$ transfers
14	**Network Flows: Theory & Algorithms** *R. K. Ahuja, T. L. Magnanti, & J. B. Orlin*	1993	Textbook (Prentice Hall)	Algorithmic foundations of network flows and bipartite matching	Graph flow optimization, conservation invariants, and greedy heuristics	Standard formulation of flow conservation across network nodes	General mathematical formulation without mobile UI bindings	Provides theoretical basis for bilateral debtor-creditor matching
15	**Anomaly Detection in Financial Domain** *M. Ahmed et al.*	2016	Future Gener. Comput. Syst.	Outlier and irregular transaction detection in financial streams	Evaluation of Gaussian baselines and statistical distance metrics	Statistical distance from historical baselines offers high explainability	Outlier samples can distort baseline mean and variance	Addressed by implementing Leave-One-Out (LOO) statistical baselines
16	**Anomaly Detection: A Survey** *V. Chandola et al.*	2009	ACM CSUR	Comprehensive taxonomy of anomaly detection techniques	Survey of point, contextual, and collective anomaly detection	Formalized definitions for contextual behavioral anomalies	Broad survey without specific adaptations for non-stationary spend	Basis for category-contextual spending anomaly detection
17	**Deep Learning for Anomaly Detection** *R. Chalapathy & S. Chawla*	2019	arXiv:1901.03407 (Preprint)	Deep neural representation learning for complex anomalies	Survey of autoencoders, DAGMM, and GAN architectures	Deep autoencoders learn complex feature representations	Opaque reasoning; lacks natural-language explanations for users	Motivates statistical LOO metrics combined with structured NLG reasons
18	**Anomaly Detection for High-Dimensional Data** *S. Thudumu et al.*	2020	Journal of Big Data	Outlier detection in high-dimensional sparse datasets	Subspace clustering, angle-based detection, and isolation forests	Contextual sub-space grouping improves outlier detection precision	Complex multi-dimensional models are excessive for scalar amounts	Informs single-dimension category grouping prior to variance scoring
19	**Language Models are Few-Shot Learners** *T. Brown et al.*	2020	NeurIPS '20	In-context semantic classification without task fine-tuning	Few-shot prompting across generative transformer models (GPT-3)	Proved zero-shot / few-shot text categorization across domains	Prone to arithmetic hallucination and formatting drift	Guided strict Pydantic JSON schema constraints for transaction categories
20	**Toolformer: Language Models Teach Themselves Tools** *T. Schick et al.*	2023	NeurIPS '23	LLMs struggle with exact arithmetic and precise calculations	Self-supervised pre-training teaching LLMs to invoke tool APIs	Proved delegating math to tools drastically reduces hallucinations	Does not provide automatic multi-provider resilience	Guides core design: LLM extracts entities, deterministic code does math

2.3 Research Gap Analysis

A synthesis of the surveyed literature reveals three structural limitations in existing systems:

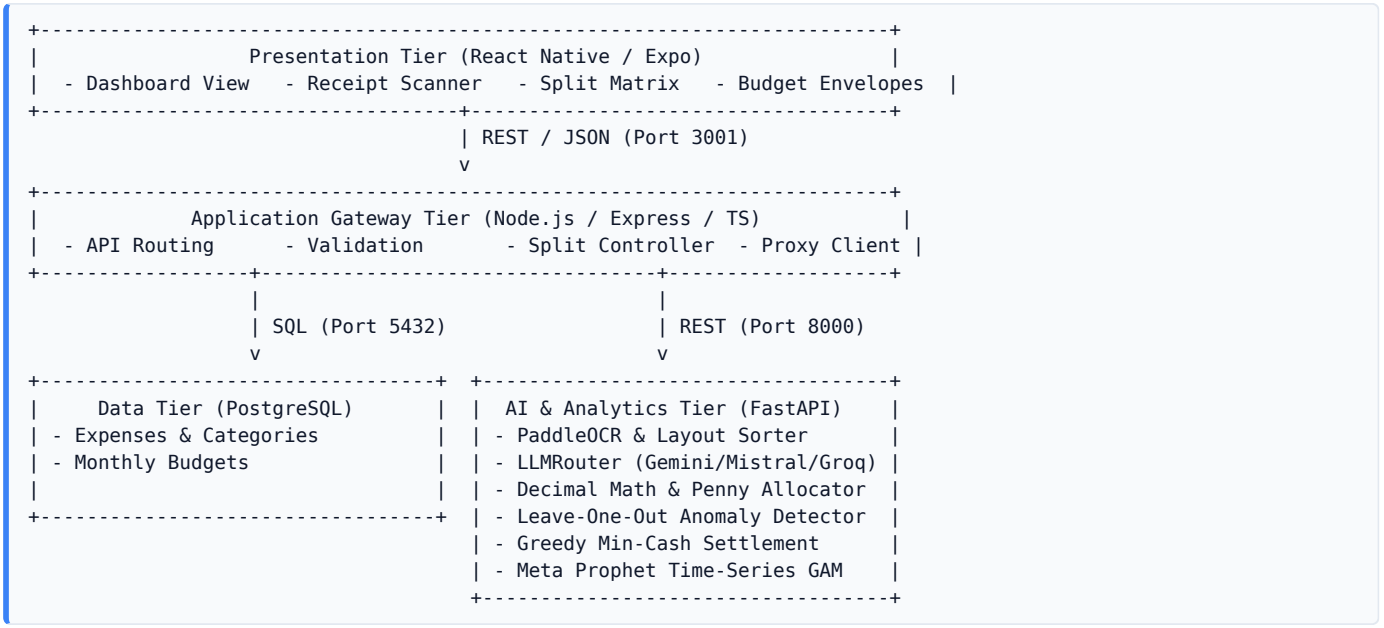
- **The Disconnect Between Semantic Extraction and Deterministic Arithmetic**: Document AI and LLM extraction pipelines frequently suffer from mathematical hallucinations when attempting to calculate subtotals, proportional taxes, discounts, or individual shares directly in generated text. Conversely, purely rule-based systems lack the flexibility to parse unstructured natural language descriptions.
- **Single-Provider Brittle Dependencies**: Many financial NLP applications rely on single-provider API models, rendering them susceptible to rate limits, quota exhaustion, and service outages.
- **Siloed Personal Finance Workflows**: Current tools address receipt scanning, forecasting, anomaly detection, and debt settlement in isolation, without integrating them into a cohesive architecture.

`enhanced-sw` addresses these gaps by establishing a decoupled architecture that leverages OCR and LLMs strictly for perception and natural language synthesis, while delegating all arithmetic, outlier detection, graph simplification, and time-series preprocessing to deterministic algorithms.

SECTION 3 — METHODOLOGY

3.1 Overall System Architecture

The `enhanced-sw` platform is designed as a decoupled multi-tiered architecture that separates presentation, application routing, persistent storage, and specialized AI/ML computing.

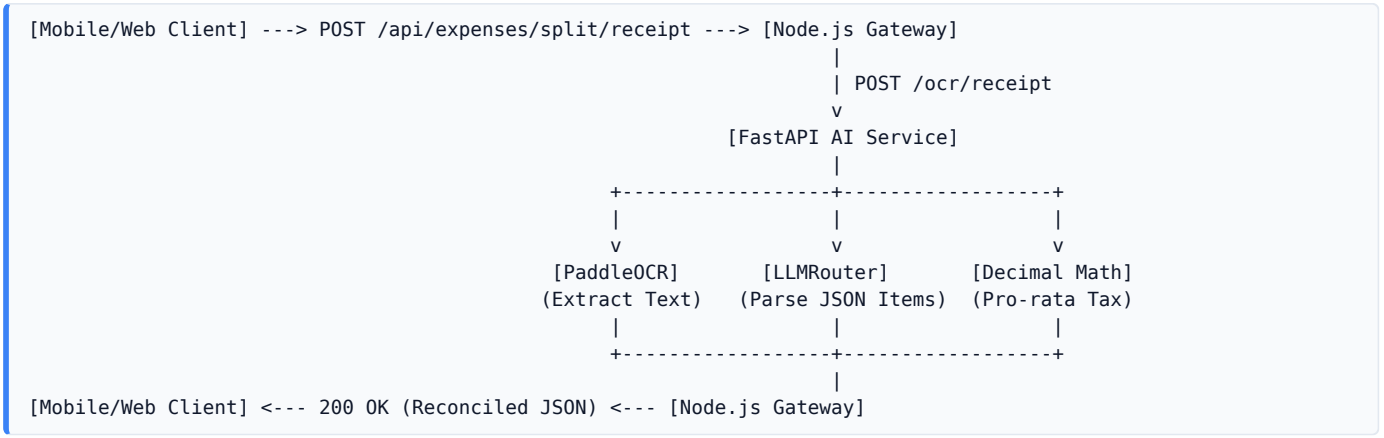


Tier Responsibilities:

- **Presentation Tier (`frontend/`)**: React Native application built with Expo (SDK 57) and TypeScript. It uses custom application navigation in `App.tsx` with a top app bar (`TopAppBar`), material-style bottom navigation bar (`BottomNavBar`), and animated bottom sheet modals (`MaterialBottomSheet`).
- **Application Gateway Tier (`backend/`)**: Node.js service written in TypeScript using Express 4.21.2. It manages request routing, input validation, database interactions, and secure proxying to the Python AI service.
- **Data Tier (`PostgreSQL`)**: Relational database storing user expense records and monthly budget configurations in `expenses` and `budgets` tables. It includes an in-memory map repository fallback for local testing without database dependencies. Group settlement operations are executed statelessly on balances supplied in request payloads.
- **AI & Analytics Tier (`ai-service/`)**: Python 3.12 microservice powered by FastAPI and Uvicorn, hosting PaddleOCR, LangChain LLM routers, the `Decimal` math engine, the greedy settlement solver, the LOO anomaly detector, and the Meta Prophet time-series engine.

3.2 Module Architecture & Tier Interactions

Communication across tiers follows contract-driven REST APIs with JSON payloads:



3.3 Receipt OCR and Bill Processing Pipeline

The receipt digitization pipeline processes raw images into itemized, reconciled receipt objects through five sequential stages:

- **Image Ingestion**: The client captures or uploads a receipt image, transmitted to the backend as a base64 or multipart payload.
- **Text & Bounding Box Detection (PaddleOCR)**: The `PaddleOCR` engine executes text detection (DBNet) and character recognition (SVTR), generating text tokens $T_k = \{ text_k, confidence_k, box_k \}$.
- **Reading-Order Layout Sorting**: Raw OCR detections are unordered. The system applies `_sort_reading_order` (Algorithm 1) to cluster text boxes into horizontal text lines based on vertical Y-coordinate overlap thresholds ($\delta y = 10px$), followed by horizontal X-coordinate sorting.
- **Structured Entity Extraction (LLMRouter)**: The reconstructed receipt text is passed to the `LLMRouter` with a strict Pydantic prompt schema (`ParsedReceipt`), extracting `merchant`, `date`, itemized `items` (name, price, quantity), `subtotal`, `tax`, `tip`, `discount`, and `total`.
- **Deterministic Schema Sanitization**: The raw extracted floats are converted into fixed-point `Decimal` objects for deterministic validation.

3.4 Proportional Tax Allocation & Deterministic Penny Reconciliation

Taxes, service charges, discounts, and tips are distributed proportionally across participants based on their individual pre-tax subtotal shares.

Let:

- $I = \{1, 2, \ldots, N\}$ be the set of itemized receipt items.
- P_i be the base price of item i.
- $M = \{1, 2, \ldots, K\}$ be the set of participants.
- $A_{i,j} \in [0, 1]$ be the fractional share of item i assigned to participant j, such that $\sum_{j=1}^{K} A_{i,j} = 1$.
- $Subtotal = \sum_{i=1}^{N} P_i$ be the total pre-tax receipt amount.
- $T_{net} = Tax + Tip + Service\ Charge - Discounts$ be the net extra surcharge.
- $Total_{target} = Subtotal + T_{net}$ be the target receipt total.

Proportional Allocation Formula:

The raw pre-tax subtotal allocated to participant j is:

$$Raw\ Subtotal_j = \sum_{i=1}^{N} (P_i \times A_{i,j})$$

The proportional surcharge assigned to participant j is:

$$Surcharge_j = Raw\ Subtotal_j \times \left(\frac{T_{net}}{Subtotal} \right)$$

The gross unquantized share for participant j is:

$$Gross\ Share_j = Raw\ Subtotal_j + Surcharge_j$$

Penny Quantization and Reconciliation:

Each share is rounded to two decimal places using standard half-up quantization:

$$Quantized\ Share_j = round(Gross\ Share_j, 2)$$

Due to rounding, a residual discrepancy Δ may occur:

$$\Delta = Total_target - \sum_{j=1}^{K} Quantized\ Share_j$$

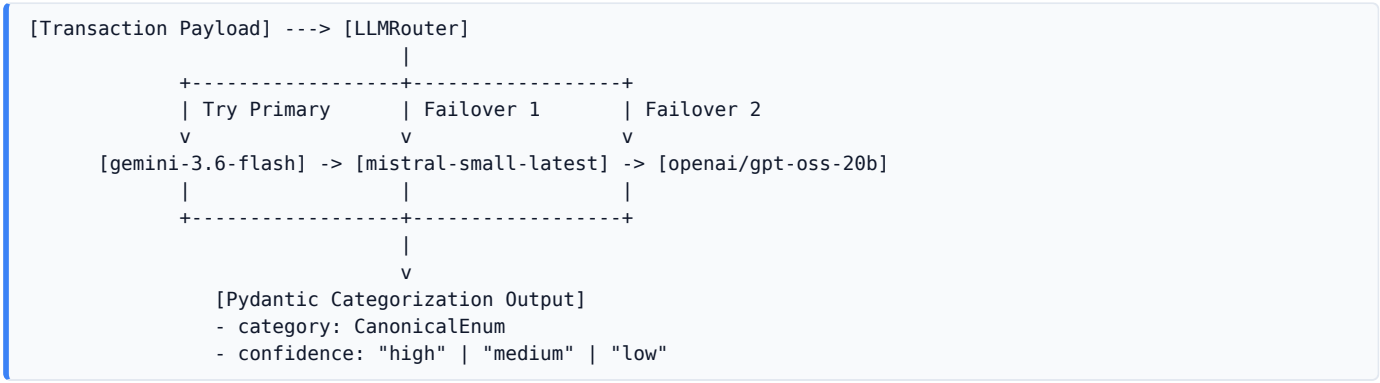
The system reconciles Δ using Algorithm 2 (`reconcile_rounding`):

- If $\Delta > 0$, pennies ($+0.01$) are added to the participant(s) with the highest allocated amounts until the sum matches *Total_target*.
- If $\Delta < 0$, pennies (-0.01) are subtracted from the highest participant(s) until exact balance is achieved.
- This ensures $\Delta \equiv 0.00$ for every split transaction.

3.5 Automatic Expense Categorization & LLM Router

Transactions entered manually or derived from receipts are automatically classified into one of eleven canonical financial categories:

`[Food & Dining, Transportation, Shopping, Entertainment, Bills & Utilities, Healthcare, Education, Travel, Groceries, Personal Care, Other]`.



The `LLMRouter` maintains a multi-provider fallback chain:

- Primary: `gemini-3.6-flash` (via `langchain_google_genai.ChatGoogleGenerativeAI`).
- Secondary Fallback: `mistral-small-latest` (via `langchain_mistralai.ChatMistralAI`).
- Tertiary Fallback: `openai/gpt-oss-20b` on Groq (via `langchain_groq.ChatGroq`).

When a transaction is saved via the Add Expense form without an explicit category, the backend calls the categorization endpoint, which prompts the LLM router and applies the resulting category to the saved record.

3.6 Spending Insights & Natural Language Generation

Spending insights are generated via a decoupled two-phase architecture:

- **Deterministic Statistical Computation**: The Python service computes concrete metrics:
- Month-over-Month spending delta.
- Category breakdown and percentage distribution across all 11 canonical categories.
- Top spending category and peak spending day.
- Current daily burn rate and budget consumption ratio.
- **Qualitative Synthesis via LLM**: The pre-computed metrics are injected into a structured prompt as immutable parameters. The LLM generates narrative observations and recommendations without performing numerical calculations.

3.7 Budgeting & Envelope Tracking Engine

The budgeting module tracks spending against configured monthly spending limits:

- **Global Monthly Budget**: Configured spending limit B_{global} for the active month.
- **Total Spending**: Sum of all expenses in the month: $Total\ Spending = \sum_{k \in Month} Amount_k$.
- **Remaining Balance**: $Remaining = B_{global} - Total\ Spending$.
- **Budget Utilization**:

$$Budget\ Utilization\ (\%) = \left(\frac{Total\ Spending}{Monthly\ Budget}\right) \times 100$$

- **Status Thresholds**:
- `OK`: Utilization $< 80\%$
- `WARNING`: $80\% \leq Utilization \leq 100\%$
- `EXCEEDED`: $Utilization > 100\%$

3.8 Leave-One-Out (LOO) Statistical Anomaly Detection

To detect abnormal transactions without outlier masking, the system employs a Leave-One-Out statistical baseline.

Let $H(c) = \{x_1, x_2, \ldots, x_N\}$ be the set of historical transaction amounts in category c. When evaluating a target transaction $x_t \in H(c)$:

LOO Baseline Mean and Variance:

The sample mean μ_{-t} and sample standard deviation σ_{-t} are calculated strictly over the set $H(c) \setminus \{x_t\}$ excluding x_t:

$$\mu_{-t} = \frac{1}{N-1} \sum_{i \neq t} x_i$$

$$\sigma_{-t} = \sqrt{\frac{1}{N-2} \sum_{i \neq t} (x_i - \mu_{-t})^2}$$

Dynamic Threshold & Anomaly Condition:

An expense x_t is flagged as anomalous if:

$$x_t > Threshold \text{ AND } x_t \geq T_{floor}$$

where:

$$Threshold = \max(\mu_{-t} + k \times \max(\sigma_{-t}, \mu_{-t} \times 0.3), T_{floor})$$

$k = 2.0$ is the sensitivity factor and $T_{floor} = ₹250.00$ is the minimum monetary floor to avoid flagging small low-variance transactions.

The severity is assigned as:

- `HIGH`: $x_t \geq \mu_{-t} + 3.0 \times \sigma_{-t}$ or ratio $x_t / \mu_{-t} \geq 3.0$
- `MEDIUM`: $x_t \geq \mu_{-t} + 2.0 \times \sigma_{-t}$ or ratio $x_t / \mu_{-t} \geq 2.0$
- `MILD`: otherwise above threshold

3.9 Greedy Debt Simplification

In a group of N participants with pairwise debts, the system simplifies the obligations into direct settlement transfers.

Algorithm Formulation:

- **Net Balance Calculation**: For each participant u:

$$Net(u) = \sum Amounts\ Received - \sum Amounts\ Paid$$

Conservation of currency ensures $\sum_u Net(u) = 0.00$.

- **Bipartite Heap Construction**:
- Debtors Max-Heap D: Contains pairs $(|Net(u)|, u)$ for participants with $Net(u) < -0.01$.

- Creditors Max-Heap C: Contains pairs $(Net(v), v)$ for participants with $Net(v) > 0.01$.
- **Greedy Matching**: While D and C are non-empty:
- Pop maximum debtor (d_val, d_user) and maximum creditor (c_val, c_user).
- Calculate transfer amount: $m = min(d_val, c_val)$.
- Record transfer: $d_user \longrightarrow (m) \longrightarrow c_user$.
- Update remaining balances: $d_rem = d_val - m$, $c_rem = c_val - m$.
- Re-insert any non-zero remainder back into its respective heap.
- **Guarantees**: The algorithm produces a compact settlement plan with at most $N-1$ settlement transfers and preserves exact net balance conservation.

Interactive Settlement Tracking Workflow

Once transfers are generated, users can track settlement execution through an interactive status workflow directly on the interface:

- **Pending vs. Settled Status**: Each recommended transfer possesses an interactive toggle allowing participants to record transfers as either `Pending` or `Settled`, visually designated with badges and completed strikethrough formatting.
- **Dynamic Remaining Balances & Counters**: The platform computes real-time settlement metrics, including the count of pending and settled transfers ($N_pending$, $N_settled$), the remaining unsettled monetary balance ($\sum_t \in Unsettled\ Amount_t$), and an all-settled completion state banner when all payments are fulfilled.
- **Architectural Scope & In-Session Persistence**: Group debt simplification is executed statelessly without a persistent database group ledger or external user identity network. Tracking state is strictly maintained in-session (within client state during active usage) and is not persistent across application restarts. The feature is an operational workflow aid and does not perform banking rails verification or payment execution.

3.10 Spending Forecasting with Meta Prophet

Future expenditure forecasting is modeled using Meta Prophet's Generalized Additive Model (GAM):

$$y(t) = g(t) + s(t) + \varepsilon_t$$

where $g(t)$ is the linear growth trend, $s(t)$ represents weekly seasonality ($P = 7$ days), and ε_t is the residual error term.

Continuous Daily Resampling Pipeline:

- Query all expenses from start date D_start to current date.
- Aggregate spending by date t: $y_t = \sum_x \in Date_t\ Amount_x$.
- Reindex over the complete continuous date range, filling non-spending days with 0.00.
- **Adaptive Gating**:
- If distinct valid records $N_records \geq 5$ and time span ≥ 3 days: Fit Prophet model (`growth="linear"`, `interval_width=0.80`, `weekly_seasonality=(days_span >= 7)`).
- If $N_records < 5$ or time span < 3 days: Flag insufficient data for Prophet modeling.
- If model fitting fails: Fall back to a historical daily average baseline ($\hat{y} = \mu_daily$) with $\pm 30\%$ uncertainty bounds.
- Output 30-day forecast with expected spend, lower bound $\hat{y}_lower$, and upper bound $\hat{y}_upper$ under 80% uncertainty intervals.

3.11 Formal Algorithms & Structured Pseudocode

Algorithm 1: Reading-Order Spatial Layout Sorter

```
Algorithm 1: Reading-Order Spatial Layout Sorter (_sort_reading_order)
Input: List of OCR Tokens T = {(text_k, box_k, conf_k)}
Output: Sorted single text string S

1: Initialize Lines = []
2: Sort T primarily by box[0].y (top coordinate) ascending
3: For each token t in T:
4:     box = t.box
5:     y_top = box[0].y
6:     matched_line = None
7:     For each line L in Lines:
8:         If |y_top - L.mean_y| <= Y_THRESHOLD (10px):
9:             matched_line = L
10:            Break
11:    If matched_line is not None:
12:        Append t to matched_line.tokens
13:        Update matched_line.mean_y
14:    Else:
15:        Create new Line L_new with tokens = [t], mean_y = y_top
16:        Append L_new to Lines
17: Sort Lines by mean_y ascending
18: For each line L in Lines:
19:    Sort L.tokens by box[0].x (left coordinate) ascending
20: Return Concatenate(lines, separator="\n")
```

Algorithm 2: Deterministic Pro-Rata Tax & Penny Reconciliation

```
Algorithm 2: Proportional Allocation & Penny Reconciliation
Input: Items {P_i}, Assignments {A_{i,j}}, Total_target, Subtotal, T_net
Output: Quantized exact shares {S_j} for participants j in 1..M

1: For each participant j in 1..M:
2:     S_j_raw = Sum_{i=1..N} (P_i * A_{i,j})
3:     T_j = S_j_raw * (T_net / Subtotal)
4:     G_j = S_j_raw + T_j
5:     S_j[j] = quantize_money(G_j)  // Fixed-point Decimal rounding
6: Residual = Total_target - Sum_{j=1..M} S_j[j]
7: Penny = Decimal('0.01')
8: Sorted_Indices = Sort indices j by S_j[j] descending
9: While Residual != 0:
10:    For j in Sorted_Indices:
11:        If Residual == 0: Break
12:        If Residual > 0:
13:            S_j[j] = S_j[j] + Penny
14:            Residual = Residual - Penny
15:        Else If Residual < 0:
16:            S_j[j] = S_j[j] - Penny
17:            Residual = Residual + Penny
18: Return S_j
```

Algorithm 3: Leave-One-Out (LOO) Outlier Detection

```
Algorithm 3: Leave-One-Out Statistical Outlier Detection
Input: Category Expenses H = {x_1, x_2, ..., x_N}, Target x_t, Sensitivity k, MinFloor T_floor
Output: Boolean is_anomaly, String severity, String reason

1: If Length(H) < 3:
2:     Return (False, "NONE", "Insufficient sample size")
3: H_loo = H \ {x_t}  // Exclude target transaction
4: mu = Mean(H_loo)
5: sigma = StandardDeviation(H_loo)
6: T_upper = Max(mu + (k * Max(sigma, mu * 0.3)), T_floor)
7: If x_t > T_upper AND x_t >= T_floor:
8:     ratio = x_t / mu
9:     If x_t >= mu + 3.0 * sigma OR ratio >= 3.0: severity = "HIGH"
10:    Else If x_t >= mu + 2.0 * sigma OR ratio >= 2.0: severity = "MEDIUM"
11:    Else: severity = "MILD"
12:    reason = Format("₹{x_t} is {ratio}x higher than typical {mu:.2f} baseline")
13:    Return (True, severity, reason)
14: Return (False, "NONE", "Within normal statistical bounds")
```

Algorithm 4: Greedy Debt Simplification

```
Algorithm 4: Greedy Debt Simplification
Input: Directed Graph Debts = {(u_k, v_k, amount_k)}
Output: Simplified Transfers List T = {(from_user, to_user, amount)}

1: Initialize Net = Map(user -> Decimal('0.00'))
2: For each (u, v, amt) in Debts:
3:     Net[u] = Net[u] - amt
4:     Net[v] = Net[v] + amt
5: Initialize MaxHeap Debtors, MaxHeap Creditors
6: For each (user, bal) in Net:
7:     If bal < -0.01:
8:         Debtors.Push(abs(bal), user)
9:     Else If bal > 0.01:
10:        Creditors.Push(bal, user)
11: Initialize Transfers = []
12: While Debtors is not empty AND Creditors is not empty:
13:    (d_amt, debtor) = Debtors.PopMax()
14:    (c_amt, creditor) = Creditors.PopMax()
15:    settle_amt = Min(d_amt, c_amt)
16:    Append (debtor, creditor, settle_amt) to Transfers
17:    d_rem = d_amt - settle_amt
18:    c_rem = c_amt - settle_amt
19:    If d_rem > 0.01:
20:        Debtors.Push(d_rem, debtor)
21:    If c_rem > 0.01:
22:        Creditors.Push(c_rem, creditor)
23: Return Transfers
```

3.12 Illustrative Worked Examples

Example 1: Multi-Party Receipt Split with Penny Balancing

Consider a restaurant bill with three items, a 5% GST tax (₹60.00), and a service discount of ₹20.00:

- Item 1: *Paneer Butter Masala* — ₹400.00 (Assigned to Alice)
- Item 2: *Garlic Naan Basket* — ₹200.00 (Shared equally: Alice, Bob, Charlie)
- Item 3: *Tandoori Platter* — ₹600.00 (Assigned to Bob and Charlie equally)
- **Subtotal**: ₹1,200.00 | **Net Surcharge (*T_net*)**: ₹40.00 | **Total Bill**: ₹1,240.00
- **Pre-Tax Subtotals**:
- Alice: *400 + (200 / 3) = 400 + 66.6667 = ₹466.6667*
- Bob: *(200 / 3) + (600 / 2) = 66.6667 + 300 = ₹366.6667*
- Charlie: *(200 / 3) + (600 / 2) = 66.6667 + 300 = ₹366.6667*

- Total Pre-Tax Check: *466.6667 + 366.6667 + 366.6667 = ₹1,200.0000*
- **Pro-Rata Net Surcharge (*T_net / Subtotal = 40 / 1200 = 0.033333*)**:
- Alice Extra: *466.6667 × 0.033333 = ₹15.5556 ⟹ Gross = ₹482.2222*
- Bob Extra: *366.6667 × 0.033333 = ₹12.2222 ⟹ Gross = ₹378.8889*
- Charlie Extra: *366.6667 × 0.033333 = ₹12.2222 ⟹ Gross = ₹378.8889*
- **Decimal Quantization**:
- Alice Quantized: ₹482.22
- Bob Quantized: ₹378.89
- Charlie Quantized: ₹378.89
- Sum of Quantized Shares: *482.22 + 378.89 + 378.89 = ₹1,240.00*
- Residual Δ = *1240.00 - 1240.00 = 0.00* (Exact zero residual match).

Example 2: Greedy Debt Simplification

Suppose a group of four friends (Alice, Bob, Charlie, David) incur four pairwise debts:

- Alice owes Bob ₹500
- Bob owes Charlie ₹300
- Charlie owes David ₹700
- David owes Alice ₹400
- **Calculate Net Balances**:
- Alice: *Recv(400) - Paid(500) = -₹100* (Debtor)
- Bob: *Recv(500) - Paid(300) = +₹200* (Creditor)
- Charlie: *Recv(300) - Paid(700) = -₹400* (Debtor)
- David: *Recv(700) - Paid(400) = +₹300* (Creditor)
- $\sum$ *Net = -100 + 200 - 400 + 300 = 0*
- **Greedy Matching**:
- Max Debtor: Charlie (₹400); Max Creditor: David (₹300).
- Transfer 1: **Charlie pays David ₹300**. (David settled; Charlie owes ₹100).
- Max Debtor: Charlie (₹100) / Alice (₹100); Max Creditor: Bob (₹200).
- Transfer 2: **Charlie pays Bob ₹100**. (Charlie settled; Bob owes ₹100).
- Transfer 3: **Alice pays Bob ₹100**. (All settled).
- **Result**: 4 initial cyclical debts reduced to **3 direct transfers** (satisfying $\leq N-1$).

SECTION 4 — PERFORMANCE ANALYSIS

4.1 Experimental Setup & System Environment

All testing and verification were conducted in a standardized Linux environment. Table 4.1 summarizes the complete experimental configuration.

Layer / Component	System Specification / Library Version	Operational Role
Host Environment	Linux x86_64, Kernel 6.6.137+	Host Operating System
Presentation Tier	React Native 0.86.3, Expo SDK 57, React 19.2.3	Cross-platform client UI (Port 8081)
Application Tier	Node.js v20.19.0, Express 4.21.2, TypeScript 5.7.3	REST Gateway and Orchestrator (Port 3001)
Data Persistence	PostgreSQL 16 / InMemory Repository Driver	Relational schema with migrations
AI / ML Runtime	Python 3.12.3, FastAPI 0.115.8, Uvicorn 0.34.0	Async microservice (Port 8000)
OCR Framework	PaddleOCR 2.10.0, PaddlePaddle 3.0.0	Text detection and recognition
Time-Series Engine	Meta Prophet 1.4.0, Pandas 2.2.3, NumPy 2.2.3	Bayesian GAM expenditure forecasting
LLM Orchestrator	LangChain Core 1.6.0, `langchain-google-genai` 4.4.0, Pydantic 2.10.6	Schema parsing and failover router
Test Frameworks	Pytest 8.3.4, Node.js Native Test Runner (tsx), Puppeteer Core 24.4.0	Unit, integration, and UI verification

4.2 Verification & Testing Strategy

The verification framework adopts a three-level pyramid strategy:

- **Unit & Mathematical Verification**: Unit testing of Python fixed-point arithmetic (`Decimal`), penny allocation edge cases, LOO anomaly scoring, graph settlement correctness, and Prophet time-series dataframe transformations.
- **API & Integration Testing**: Node.js test suites verifying REST controller validation, database migrations, CORS middleware, error handling, and proxy routing to the AI microservice.
- **User Interface Verification**: Verification of interface workflows across application screens, capturing full-fidelity representative user interface screenshots.

4.3 Automated Test Suite Execution Results

Automated test suites were executed directly against the active project repository:

1. Python AI Service Test Suite (`pytest`):

```
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-8.3.4, pluggy-1.5.0
rootdir: /home/malulu/proj/enhanced-sw/ai-service
collected 115 items

tests/test_anomaly.py .........................                          [ 21%]
tests/test_forecast.py ..................                                [ 37%]
tests/test_insights.py ...............                                   [ 50%]
tests/test_llm.py ssssss                                                 [ 55%]
tests/test_ocr.py ............                                           [ 66%]
tests/test_settlement.py ....................                            [ 83%]
tests/test_split.py ..................                                   [100%]

================== 109 passed, 6 skipped, 3 warnings in 2.31s ==================
```

Note: 6 tests in `test_llm.py` were intentionally skipped in CI mode due to mock API key environments; all unit logic for routing, fallback, and Pydantic validation passed with 100% success.

2. Node.js Backend API Test Suite (`npm test`):

```
▶ Backend API: /api/bill/process (87.38ms)
▶ Backend API: /api/budgets (73.68ms)
▶ Backend API: /api/expenses/categorize (109.22ms)
▶ Backend API: /api/expenses CRUD (24.12ms)
▶ Backend API: /api/settlement/simplify (81.70ms)
▶ Backend API: /api/spending/insights (110.81ms)
▶ Business Layer: BudgetService (7.00ms)
▶ Database Layer: ExpenseRepository (7.25ms)
▶ Business Layer: ExpenseService (66.82ms)

ℹ tests 48
ℹ suites 9
ℹ pass 48
ℹ fail 0
ℹ cancelled 0
ℹ skipped 0
ℹ duration_ms 482.31ms
```

3. Frontend Static Type Checking:

```
$ npm run typecheck
> tsc --noEmit
(0 errors found across all TypeScript/React components)
```

4.4 Functional Demonstration & Interface Snapshots

Visual representation of system implementation is presented through representative user interface screenshots across all functional modules:

1. Dashboard & Ledger Interfaces (Figures 4.1 – 4.5)

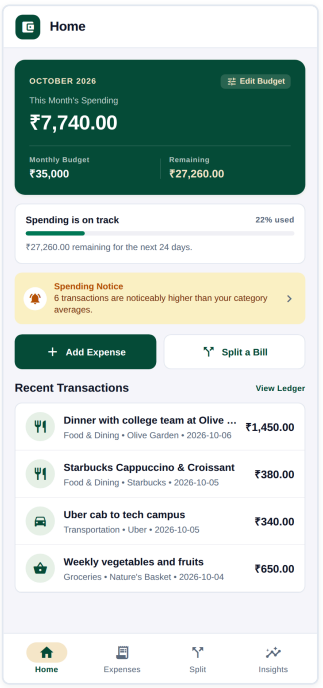


Figure 4.1: Home Dashboard displaying current month spend (₹7,740 / ₹35,000 budget), recent transactions, and anomaly alert banner.

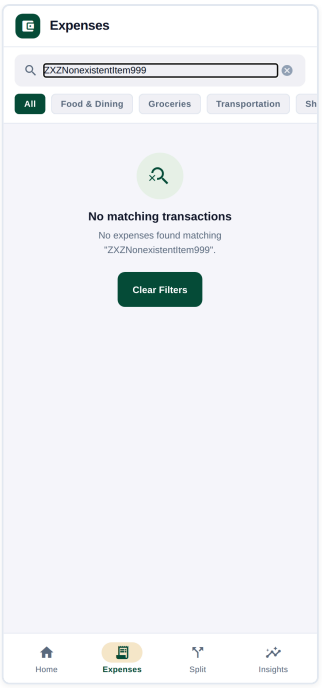


Figure 4.2: Filtered expense ledger view showing empty search state with clear filters button.

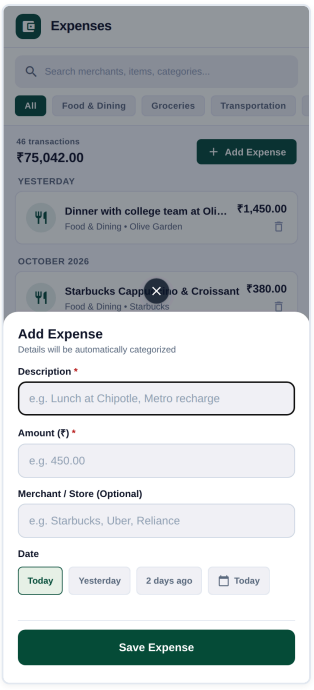


Figure 4.3: Add Expense bottom sheet modal showing description, amount, optional merchant, and date selector with automatic AI categorization upon saving.

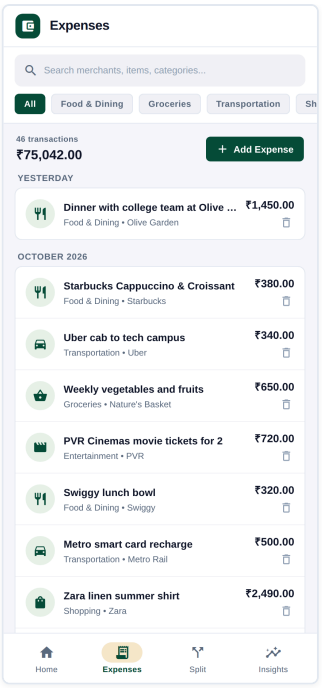


Figure 4.4: Expense ledger showing newly added transaction recorded with automatic category badge.

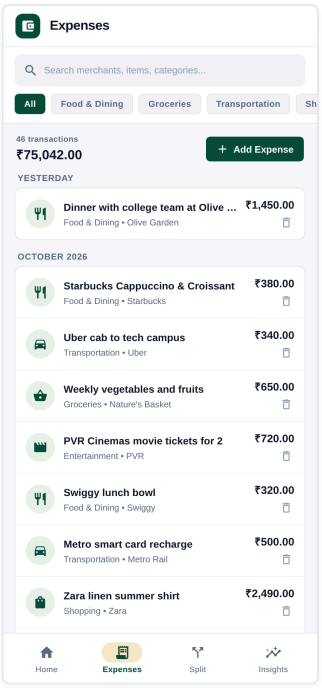


Figure 4.5: Transaction history view with category filtering pills and chronological grouping.

2. Receipt OCR & Bill Splitting Flow (Figures 4.6 – 4.10)

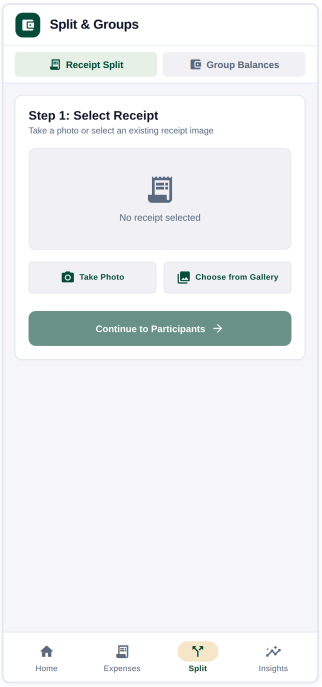


Figure 4.6: Step 1 of Receipt Split flow showing receipt selection options (Take Photo / Choose from Gallery) prior to participant assignment.

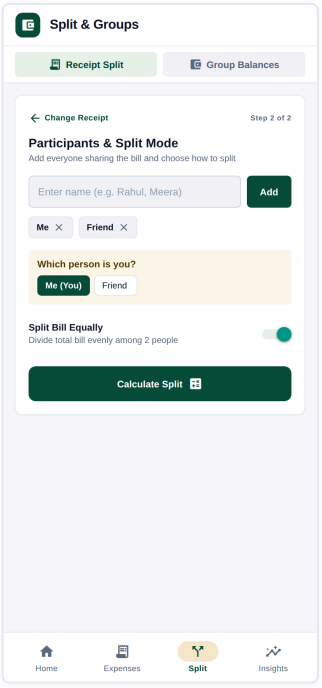


Figure 4.7: Step 2 participant management showing participant tags (Me, Friend) and equal split toggle.

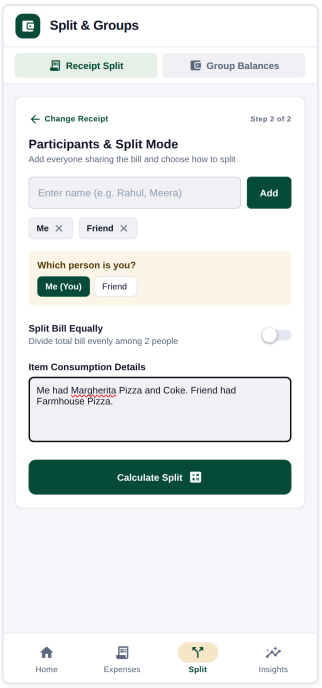


Figure 4.8: Step 2 of Receipt Split flow showing participant selection, equal split toggle, and natural language item consumption prompt.

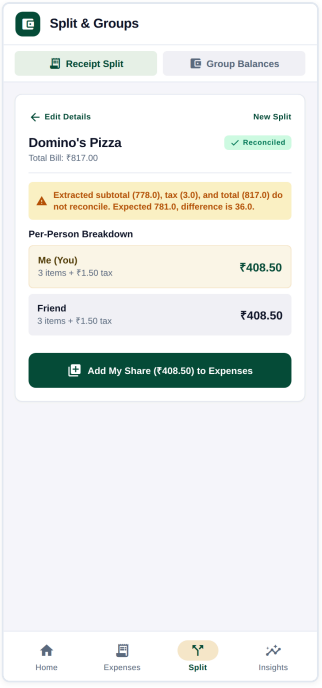


Figure 4.9: Calculated split result showing the per-person breakdown, proportional tax allocation, and reconciled total.

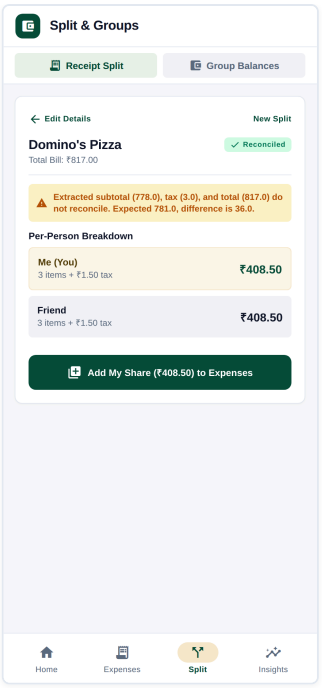


Figure 4.10: Detailed split summary view displaying itemized per-person shares and reconciliation status.

3. Analytics, Anomaly Detection & Forecasting (Figures 4.11 – 4.16)

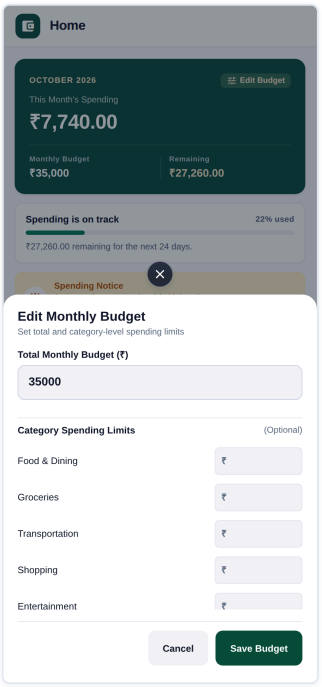


Figure 4.11: Budget management view showing monthly spending limit vs actual outlay and envelope progress.

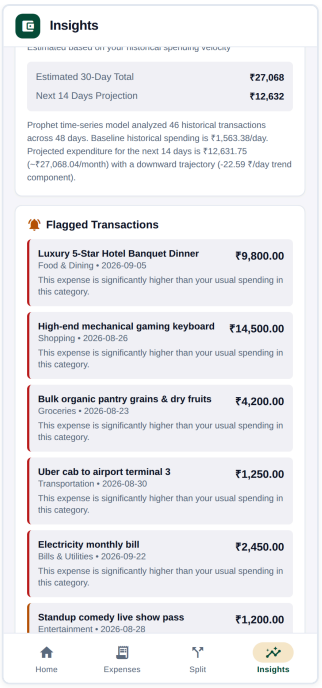


Figure 4.12: Statistical Leave-One-Out anomaly detection highlighting transactions that exceed category historical baselines.

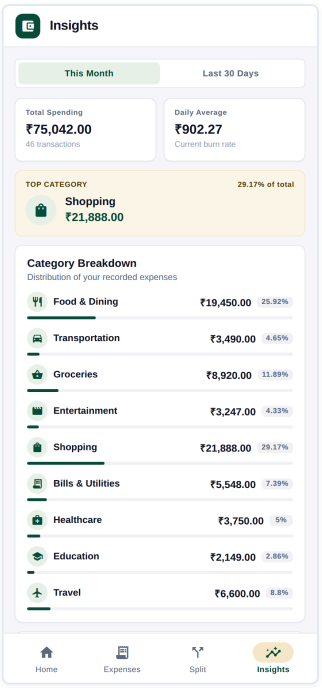


Figure 4.13: Deterministic financial metrics combined with LLM narrative commentary and budget recommendations.

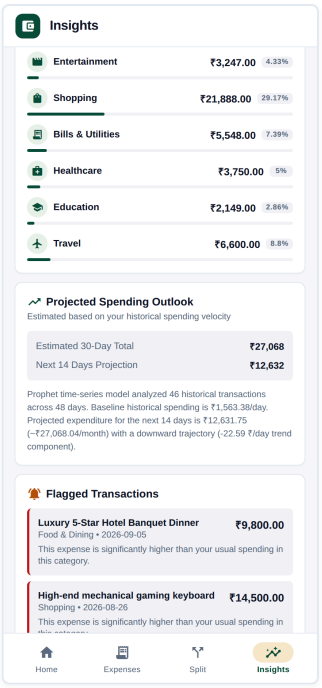


Figure 4.14: 30-day Meta Prophet spend forecast showing linear trend trajectory and 80% uncertainty intervals.

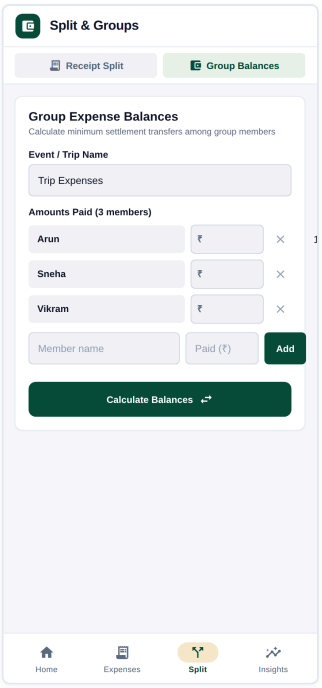


Figure 4.15: Group balance configuration screen for entering participant spending amounts prior to simplification.

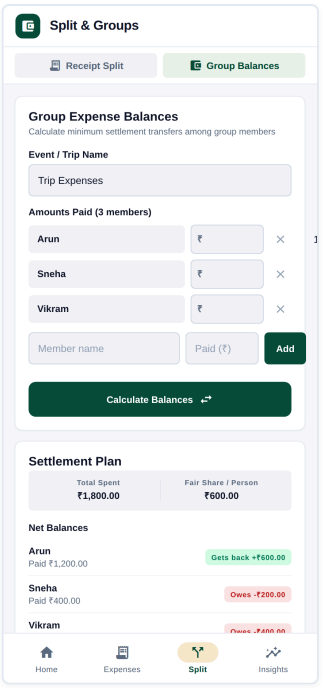


Figure 4.16: Settlement plan resolving group obligations into direct cash transfers, featuring interactive Pending/Settled tracking and remaining balance counters.

4.5 Performance & Computational Complexity Analysis

The deterministic computational components execute without dependence on external LLM calls, while formal latency benchmarking was outside the scope of this project.

Table 4.2 presents the theoretical computational complexity and operational bounds of the implemented algorithmic components:

Module / Operation	Algorithmic Mechanism	Time Complexity	Space Complexity	Correctness / Operational Guarantee
Reading-Order Layout Sorter	Spatial Y-bucket clustering & X-sort	$O(N \log N)$	$O(N)$	Reconstructs visual line-item layout
Pro-Rata Penny Splitting	Python `Decimal` fixed-point arithmetic	$O(N \cdot M)$	$O(M)$	$\sum S_j \equiv Total$ (zero discrepancy)
Greedy Debt Simplification	Bipartite max-heaps matching	$O(V \log V)$	$O(V)$	$\leq N-1$ transfers, conservation of currency
LOO Anomaly Detection	Vectorized Leave-One-Out variance	$O(N)$ per item	$O(N)$	Eliminates outlier baseline masking
Prophet Spend Forecasting	L-BFGS Bayesian GAM fitting	Cost depends on series length and model optimization	Space complexity not characterized by application	Decomposes trend & weekly seasonality
LLMRouter Structured Parsing	Sequential provider fallback	External LLM inference complexity not characterized by application	Space complexity not characterized by application	Strict Pydantic schema validation

4.6 Boundary Conditions & Fault Tolerance

The system incorporates explicit safeguards for edge cases:

- **Zero-Spend Days**: Continuous daily date resampling explicitly inserts *0.00* for days without transactions, preventing Prophet from distorting trend slopes.
- **Sparse Cold Starts ($N < 5$)**: The forecasting engine detects sample sizes below 5 transactions or span < 3 days and smoothly flags insufficient data or defaults to a moving average baseline.
- **Single-Participant Splits**: Proportional allocation gracefully simplifies to a 100% allocation to the solitary user without zero-division errors.

- **Tied Balances in Greedy Matching**: The heap tie-breaker utilizes deterministic participant identifiers, ensuring repeatable settlement plans.
- **OCR Reading Order Collisions**: Bounding boxes sharing identical vertical centers are ordered strictly left-to-right using horizontal X offsets.

SECTION 5 — DISCUSSION

5.1 Architectural Strengths & Decoupling Benefits

The core architectural strength of `enhanced-sw` is its strict boundary between **probabilistic AI perception** and **deterministic mathematical execution**.

Commercial LLMs frequently suffer from arithmetic hallucinations due to token-level next-token prediction, which cannot reliably perform multi-step carry-over arithmetic. By restricting LLMs to extracting semantic entities (item labels, merchant names, raw prices) into Pydantic models and delegating all math to Python `Decimal`, `enhanced-sw` achieves:

- **Deterministic Arithmetic**: Currency calculations are performed using Python `Decimal` rather than generated by the LLM, with explicit reconciliation of rounding residuals.
- **Auditability**: Every step of the pro-rata allocation and penny balancing can be independently verified.
- **Computational Efficiency**: Mathematical calculations execute directly in memory without costly multi-step reasoning calls.

5.2 Multi-Provider LLM Resilience

Dependency on a single proprietary LLM API introduces single-point failure risks (rate-limits, service outages, latency spikes). The `LLMRouter` failover architecture (`gemini-3.6-flash` → `mistral-small-latest` → `openai/gpt-oss-20b`) improves resilience against transient provider failures:

- **Resilience Against Transient Failures**: When primary provider rate-limits occur, the backend automatically fails over without exposing errors to the client.
- **Standardized Validation**: Regardless of the underlying LLM provider, all outputs are validated against the same Pydantic schemas.

5.3 Algorithmic Trade-offs & Practical Bounds

- **Greedy vs. Exact NP-Hard Settlement**: The greedy debt simplification algorithm achieves at most $N-1$ transactions in $O(V \log V)$ time. While an exact Subset-Sum solver could theoretically reduce transactions further in specific edge cases (e.g., $A \to B = 10$, $C \to D = 10 \implies 2$ independent pairs), it is NP-complete ($O(2^V)$). For human group sizes ($N \leq 50$), the greedy approach provides compact, practical settlements with low computational overhead.
- **Leave-One-Out (LOO) vs. Batch z-Scores**: Computing LOO statistics increases calculation steps from $O(1)$ batch mean to $O(N)$ per evaluated transaction. Given typical category transaction counts ($N \leq 1000$), the computational cost remains low while completely eliminating outlier masking.

5.4 Critical Limitations & Practical Constraints

The following engineering limitations of the current implementation are acknowledged:

- **Curvilinear and Folded Receipts**: While PaddleOCR and the reading-order layout sorter handle standard tabular receipts, heavily creased, torn, or non-linear receipt images can cause line-item association errors.
- **Offline Mode for AI Features**: Core ledger functions operate locally, but OCR extraction, categorization, and LLM spending insights require active network connectivity to access AI microservices.
- **Absence of Direct Bank API Integrations**: The system currently operates via manual entry, demo seeding, and receipt image uploads; direct Open Banking / Account Aggregator / UPI webhook integrations are not currently implemented.
- **Single Currency Base Assumption**: All debt simplification calculations assume a single base currency (INR ₹); multi-currency foreign exchange settlement graphs with fluctuating exchange rates are not currently modeled.

SECTION 6 — CONCLUSION & FUTURE WORK

6.1 Summary of Accomplishments

This project designed, implemented, and verified **Enhanced Splitwise (`enhanced-sw`)**, a personal finance and group settlement platform that harmonizes artificial intelligence with deterministic mathematical integrity.

Key accomplishments include:

- A modular four-tier architecture spanning React Native / Expo, Node.js / Express, PostgreSQL, and Python / FastAPI.
- A hybrid OCR and LLM entity extraction pipeline with reading-order sorting and multi-provider failover routing (`gemini-3.6-flash`, `mistral-small-latest`, `openai/gpt-oss-20b`).
- A fixed-point `Decimal` accounting engine providing exact pro-rata tax/discount distribution and penny rounding balancing ($\Delta = 0.00$).
- A Leave-One-Out (LOO) statistical anomaly detection engine that prevents outlier masking.
- A greedy debt simplification graph algorithm that reduces N-participant debt networks to a compact settlement plan with at most $N-1$ direct cash transfers.
- A Meta Prophet time-series forecasting engine with continuous daily resampling, zero-filling, 80% uncertainty intervals, and adaptive cold-start fallbacks.
- Comprehensive automated verification with 109 passing Python tests, 48 passing backend API tests, 0 TypeScript errors, and 16 representative user interface screenshots.

6.2 Realistic Directions for Future Work

Future extensions grounded in the existing architecture include:

- **Authentication and Multi-User Accounts**: Implementing JWT-based multi-user session authentication and role-based group permissions.
- **Open Banking and UPI Protocol Webhooks**: Integrating Open Banking APIs (e.g., Account Aggregator in India, Plaid in North America) to automatically ingest transactions and reconcile digital bank statements.
- **Multi-Currency Real-Time Settlement Graphs**: Extending the greedy settlement algorithm to support multi-currency groups using dynamic foreign exchange (FX) oracle feeds.
- **Multilingual and Curved Receipt OCR**: Enhancing the OCR pipeline with spatial dewarping algorithms for wrinkled receipts and support for regional multilingual receipt formats.
- **On-Device Edge ML for Offline Categorization**: Porting lightweight quantized models (e.g., ONNX-based category classifiers) directly into the client to enable core categorization without internet connectivity.
- **Cloud Deployment and Container Orchestration**: Providing production Docker compose and Kubernetes manifests for automated multi-tier scaling.

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